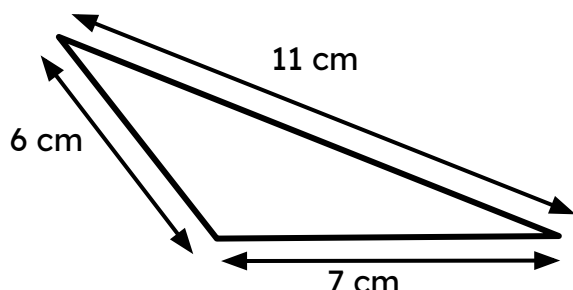




Constructing triangles

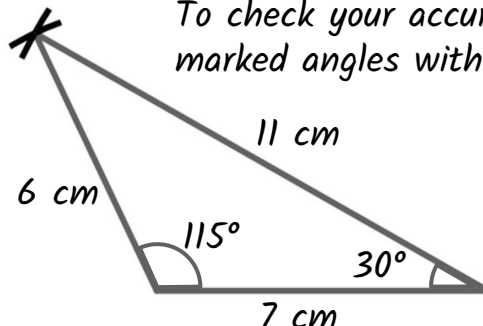
Task A: Constructing a scalene triangle

1) Construct this triangle accurately.

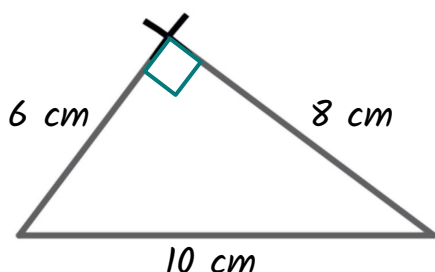


Not drawn to scale.

To check your accuracy, measure the marked angles with a protractor.



2) Construct the triangle with edge lengths 6 cm, 8 cm and 10 cm.



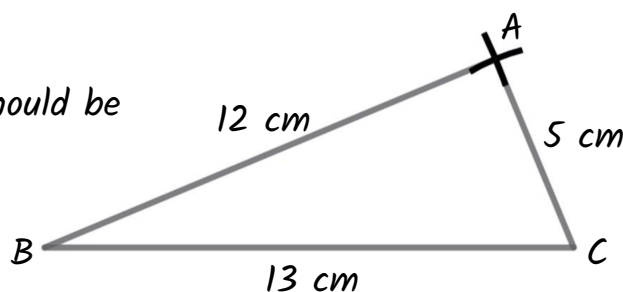
Not drawn to scale.

The angle between the 6 cm edge and the 8 cm edge should be 90° . Use a protractor to check it.

3) Construct triangle ABC, where $AB = 12$ cm, $BC = 13$ cm and $AC = 50$ mm.

Not drawn to scale.

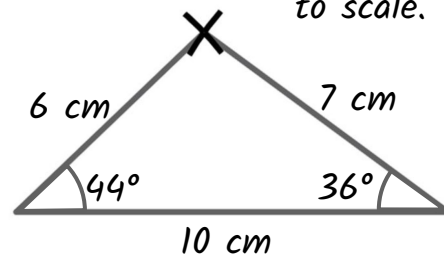
The angle between AB and AC should be 90° . Use a protractor to check it.



4) Construct the scalene triangle that has an edge of 6 cm, an edge of 7 cm and a perimeter of 23 cm.

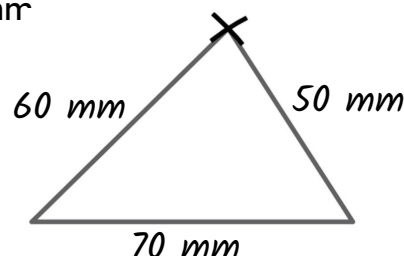
If the perimeter is 23 cm and two of the edges sum to 13 cm, then the third edge will need to have a length of 10 cm. To check your accuracy, measure the marked angles with a protractor.

Not drawn to scale.



5) Construct a scalene triangle with a perimeter of 180 mm

An example of a correct answer
(Not drawn to scale):



Task B: Constructing triangles with equal edges

1) XY is an edge of a triangle. Which of the points A to G when joined to XY would create:

a) an isosceles triangle

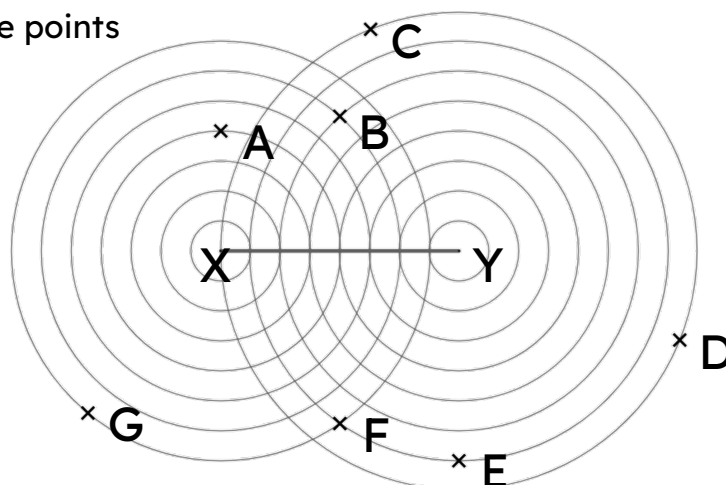
B, C, D, F

b) a right-angled triangle

A and E

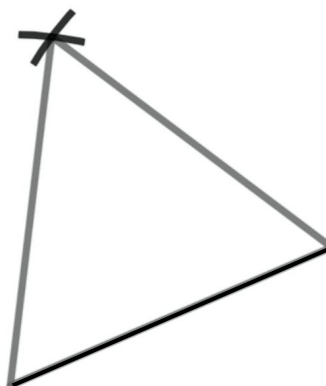
c) a right-angled isosceles triangle

There isn't one.



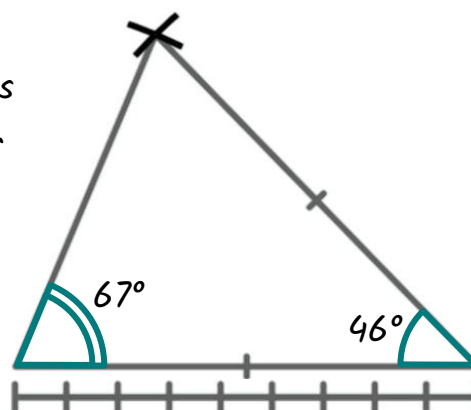
2) Construct an equilateral triangle, using the line as one of the edges.

Check your angles with a protractor; they should all be 60°



3) Construct an isosceles triangle where the equal edges are 9 units and the base edge is 7 units. Use this scale.

Check your angles with a protractor.



4) A tetrahedron is a 3D shape with four equilateral triangular faces. Construct the net of a tetrahedron. Draw this on a separate sheet of paper.

Two distinct nets of a tetrahedron

